

IT2214 Predictive Analytics Report

Name: Kahlen Ong

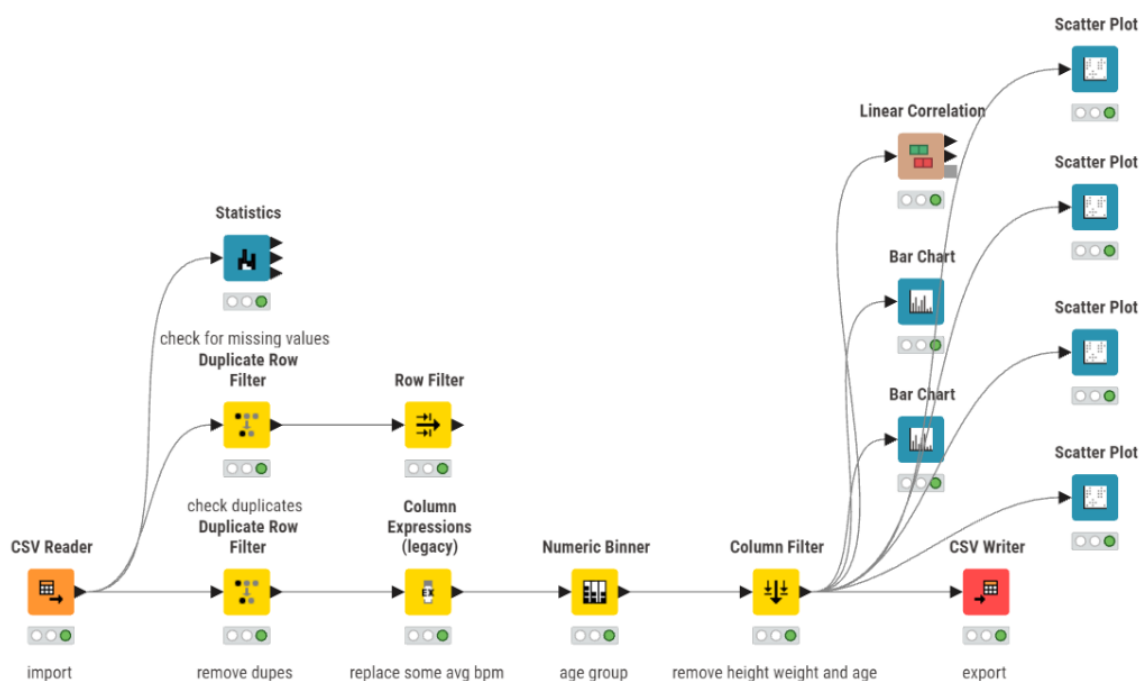
Admin Number: 252642M

Chapter 1: Analysis Objective and Workflow

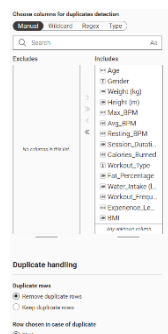
The analysis addresses the gym operator's requirement to identify how demographic and physiological factors are associated with Calories_Burned and to develop a predictive model for estimating calories burned during each workout session. The target variable is therefore Calories_Burned, with predictors representing heart rate, workout duration, body composition, workout behaviour, and demographic characteristics.

The workflow consisted of data preparation and exploratory analysis in KNIME, followed by predictive modelling and evaluation in SAS Viya. Four regression models were developed: Linear Regression, Decision Tree, Random Forest, and Gradient Boosting. A common training-validation-test partition was used to ensure that model performance could be compared fairly.

Chapter 2: Data Preparation & Exploration



The original dataset contained 4865 observations. The dataset was first profiled using the Statistics node to identify data quality issues. No missing values were detected across the variables, so no imputation was required.

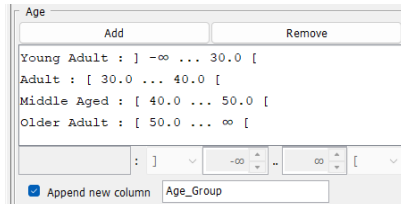


Exact duplicates were identified by comparing all columns using the Duplicate Row Filter. A total of 3892 duplicate rows were removed, leaving 973 unique observations. Removing only fully identical rows ensured that legitimate observations with similar characteristics were retained. Exact duplicates would otherwise cause repeated records to contribute disproportionately to summary statistics, correlations, and model estimation. A separate diagnostic branch in the KNIME workflow was used to verify the duplicate removal process and did not constitute a second cleaning step.

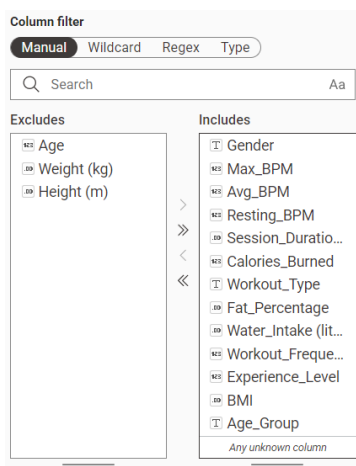
```
if (column("Avg_BPM") > column("Max_BPM")) {  
  column("Max_BPM")  
} else {  
  column("Avg_BPM")  
}
```

A logical check identified 23 observations where Avg_BPM > Max_BPM. Since an average workout heart rate should not exceed the recorded maximum heart rate,

these values were treated as inconsistent. Instead of deleting the 23 observations, Avg_BPM was replaced with the corresponding Max_BPM using the Column Expressions node. The correction assumes that these members were exercising at their recorded maximum heart rate. Replacement was preferred because the dataset had already been reduced substantially from 4865 to 973 unique observations. Capping Avg_BPM at Max_BPM retained the observations while enforcing the logical condition: $\text{Avg_BPM} \leq \text{Max_BPM}$. This assumption is acknowledged because the true erroneous average heart-rate values are unknown.



Age was transformed into an Age_Group categorical feature using the Numeric Binner node, where you can see the different groups in the picture on the left. This allows age-related differences to be interpreted as demographic groups rather than year-by-year changes.



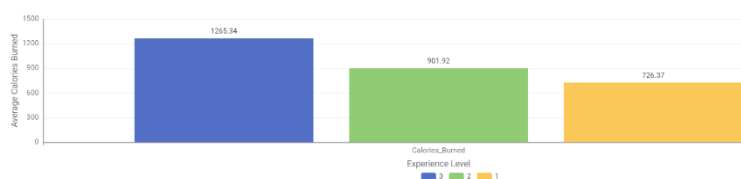
After Age_Group was created, the original Age variable was removed using the Column Filter node to avoid representing the same demographic characteristic twice. Weight (kg) and Height (m) were also removed using the same node while BMI was retained. BMI is derived from weight and height, so retaining all 3 variables would introduce overlapping information and increase the risk of multicollinearity, particularly in Linear Regression. Retaining BMI preserves a composite measure of body size while reducing redundant predictors.

The final modelling dataset therefore contained 973 observations with 13 columns:

Target: Calories_Burned; Continuous Predictors: Max_BPM, Avg_BPM, Resting_BPM, Session_Duration (hours), Fat_Percentage, Water_Intake (liters), Workout_Frequency (days/week), BMI; Categorical Predictors: Gender, Workout_Type, Experience_Level, Age_Group

Six visualisations were used to examine how selected predictors relate to Calories_Burned.

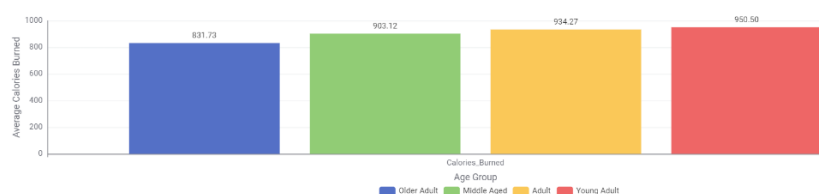
Average Calories Burned by Experience Level



The **Average Calories Burned by Experience Level** bar chart showed a clear increase in average calorie expenditure as experience level increased.

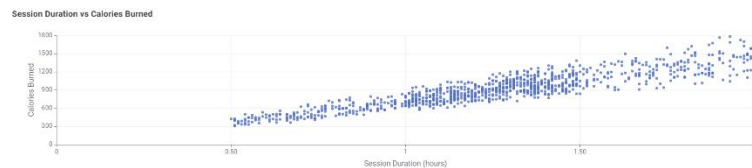
Hence, this indicates a strong association between Experience_Level and Calories_Burned. However, this should not be interpreted as a causal relationship because more experienced members may also differ in workout duration, intensity, or other physiological characteristics.

Average Calories Burned by Age Group

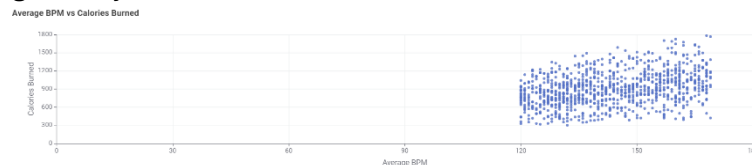


The **Average Calories Burned by Age Group** bar chart showed a gradual decrease in average calorie expenditure across older age groups.

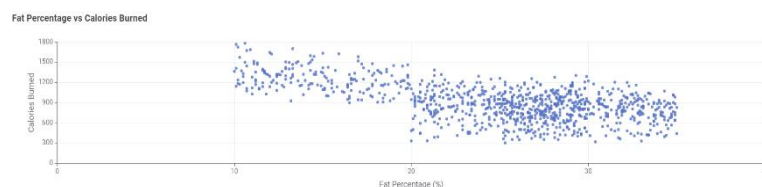
The differences are smaller than those seen in the Experience Level bar chart, which suggests that Age_Group is associated with calorie expenditure, but the relationship appears weaker than the differences observed across experience levels.



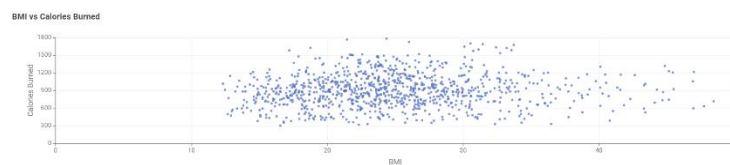
continuous variables examined. As Session_Duration (hours) increased, Calories_Burned generally increased as well.



values were generally associated with higher calorie expenditure, but members with similar average heart rates could still record substantially different calorie values.



with lower Fat_Percentage, while members with higher fat percentages generally recorded lower calorie values. However, substantial overlap remained across the observations, so Fat_Percentage contributes supporting predictive information rather than acting as a strong standalone predictor.



weak standalone predictive strength compared with variables such as Session_Duration.

Linear Correlation

Column selection
☒ Manual ☐ Wildcard ☐ Regex

Search: Aa

Excludes: ☒ Gender ☒ Workout_Type ☒ Age_Group

Includes: ☒ Max_BPM ☒ Avg_BPM ☒ Resting_BPM ☒ Session_Duration ☒ Calories_Burned ☒ Fat_Percentage ☒ Water_Intake (lit...) ☒ Workout_Frequency ☒ Experience_Level ☒ BMI ☐ Any unknown column

Output column pairs
☐ All column pairs
☐ Only column pairs of compatible columns
☒ Only column pairs with a valid correlation

Possible values count:

p-value
☒ two-sided

The Linear Correlation node was used to assess pairwise linear relationships among the numeric values shown in the screenshot. Gender, Workout_Type, and Age_Group were excluded from the correlation analysis as they were categorical values. Session_Duration had the strongest correlation with Calories_Burned ($r = 0.908$), followed by Experience_Level ($r = 0.694$). Fat_Percentage showed a negative relationship ($r = -0.598$), while Workout_Frequency had a moderate positive relationship ($r = 0.576$). Water_Intake ($r = 0.357$) and Avg_BPM ($r = 0.343$) were weaker, whereas BMI, Resting_BPM, and Max_BPM showed negligible correlations. Strong relationships between predictors, such as Workout_Frequency-Experience_Level ($r = 0.837$) and Session_Duration-Experience_Level ($r = 0.765$), further support using multivariable predictive models.

Chapter 3: SAS Viya Modelling and Comprehensive Test Design

Before model development, each variable was assigned according to its analytical role.

- Category
 - Age_Group - 4
 - Experience_Level - 3
 - Gender - 2
 - Partition - 3
 - Workout_Type - 4
- Measure
 - Avg_BPM
 - BMI
 - Calories_Burned
 - Fat_Percentage
 - Frequency
 - Max_BPM
 - Resting_BPM
 - Session_Duration (hours)
 - Water_Intake (liters)
 - Workout_Frequency (days/...

Calories_Burned was defined as the continuous response variable. Max_BPM, Avg_BPM, Resting_BPM, Session_Duration, Fat_Percentage, Water_Intake, Workout_Frequency, and BMI were treated as continuous predictors. Gender, Workout_Type, Age_Group, and Experience_Level were treated as categorical predictors. Although Experience_Level is numerically coded from 1 to 3, its values represent ordered experience categories rather than measurements on a continuous scale. Treating it as categorical prevents the models from assuming that the numerical difference between Levels 1 and 2 is quantitatively identical to the difference between Levels 2 and 3.

The cleaned dataset was then partitioned into 60% Training / 20% Validation / 20% Test. The same partition was retained across all 4 candidate models to ensure fair performance comparison.

The training partition was used to fit the models. The validation partition was used for hyperparameter selection and model refinement. The test partition was used as the final assessment of generalisation after model settings had been selected. Validation ASE was used as the primary tuning criterion for the nonlinear models. Lower Validation ASE indicates lower average squared prediction error on observations not used directly to fit the model.

Chapter 3.1: Linear Regression

Linear Regression was developed as an interpretable baseline for assessing the overall relationship using all justified continuous and categorical predictors.



Unlike the tree-based models, the Linear Regression model was not subjected to hyperparameter tuning. Model quality was instead assessed through fit statistics and residual diagnostics.

The model achieved: $R^2 = 0.9852$, Adjusted $R^2 = 0.9847$, Training ASE $\approx 1.1k$, Validation ASE $\approx 1.5k$, Test ASE $\approx 1.1k$. The high R^2 indicates that the model explains a large proportion of variation in Calories_Burned. However, the residual plot showed a noticeable curved pattern rather than purely random residual dispersion, indicating that the predictor-target relationships are not completely linear despite the strong overall model fit. The residual behaviour therefore justified evaluating nonlinear tree-based models capable of modelling more complex relationships.

Chapter 3.2: Decision Tree

Decision Tree was selected to capture nonlinear relationships and threshold-based interactions between predictors while retaining an interpretable model structure. The initial Decision Tree used: **Maximum Branches = 2, Maximum Levels = 6, Leaf Size = 5, Predictor Bins = 50**. The initial model achieved approximately: **Training ASE $\approx$ 4.2k, Validation ASE $\approx$ 7.6k, Test ASE $\approx$ 7.4k**. Tree complexity and predictor binning were subsequently refined using Validation ASE. Maximum depth, terminal leaf size and predictor bins were varied while validation pruning was used to limit unnecessary tree complexity.



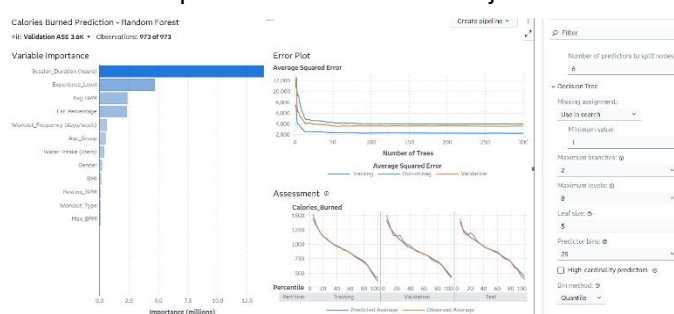
The final Decision Tree configuration was: **Maximum Levels = 8, Leaf Size = 5, Maximum Branches = 2, Predictor Bins = 100, Bin Method = Quantile, Prune with validation data = enabled**.

The final model produced: **Training ASE $\approx$ 2k, Validation ASE $\approx$ 5.2k,**

Test ASE $\approx$ 6.1k. The reduction in both validation and test error demonstrates that tuning materially improved the model. However, the final Test ASE remained higher than Validation ASE, indicating weaker generalisation than the ensemble approaches. Session_Duration (hours) was the dominant predictor, followed by Avg_BPM.

Chapter 3.3: Random Forest

Random Forest was selected to improve the stability and generalisation of a single Decision Tree by combining predictions from multiple randomised trees. The initial Random Forest used: **Number of Trees = 100, Maximum Levels = 5, Leaf Size = 5, Predictor Bins = 50, and the default number of predictors considered per split**. The baseline model achieved approximately: **Training ASE $\approx$ 5.3k, Validation ASE $\approx$ 6.8k, Test ASE $\approx$ 7.4k**. The model was then refined by testing the number of trees, predictors considered per split, maximum depth, leaf size, and predictor bins. Increasing the number of trees improved ensemble stability, while controlling the number of predictors considered at each split introduced more diversity among trees. Tree depth and leaf size were adjusted to balance model flexibility and generalisation.



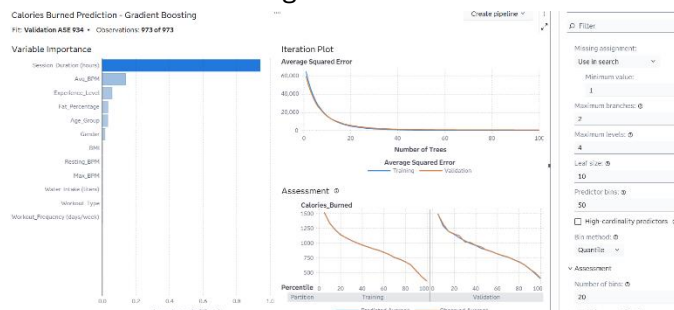
The final Random Forest configuration was: **Number of Trees = 300, Maximum Levels = 8, Leaf Size = 5, Predictor Bins = 25, Predictors Considered Per Split = 6, Bin Method = Quantile**.

The final model achieved: **Training ASE $\approx$ 2.3k, Validation ASE $\approx$ 3.6k, Test ASE $\approx$**

3.9k. The relatively small difference between Validation and Test ASE indicates reasonably stable generalisation. The Error Plot also showed training, out-of-bag, and validation error declining rapidly before stabilising as the number of trees increased. This indicates that the ensemble reached a point where additional trees provided limited further improvement. Session_Duration was again the strongest predictor, followed by Experience_Level, Avg_BPM, and Fat_Percentage.

Chapter 3.4: Gradient Boosting

Gradient Boosting was selected to capture complex nonlinear relationships by sequentially fitting trees that focus on errors made by earlier trees. The initial Gradient Boosting configuration was: **Number of Trees = 50, Learning Rate = 0.1, Subsample Rate = 0.5, Maximum Levels = 5, Leaf Size = 5, Predictor Bins = 50**. The initial model produced: **Training ASE = 349, Validation ASE $\approx$ 1.2k, Test ASE = 950**. Although the baseline model already performed strongly, further tuning was conducted to determine whether validation performance could be improved. Moving from 50 to 100 trees reduced Validation ASE from approximately 1.2k to 1.0k. Increasing the ensemble to 200 trees did not materially improve Validation ASE, while Training ASE continued to decrease. Therefore, 100 trees was retained. Tree complexity was refined by testing alternative maximum-level and leaf-size combinations. A configuration of 4 maximum levels and leaf size 10 achieved the strongest validation performance while maintaining a relatively compact tree structure. Predictor bin settings were then compared, and since the 50-bin configuration achieved the lowest Validation ASE, it was therefore retained.

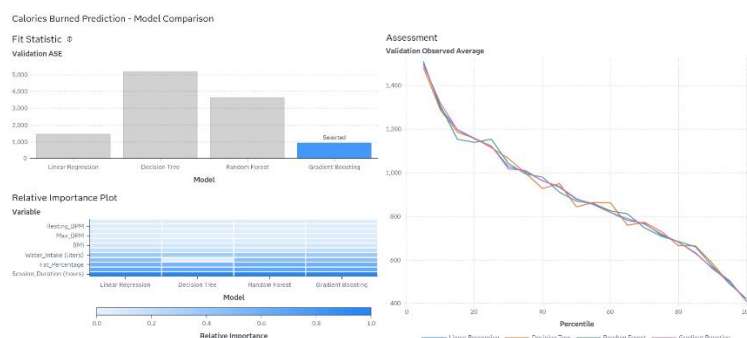


The final Gradient Boosting configuration was: **Number of Trees = 100, Learning Rate = 0.1, Subsample Rate = 0.5, Maximum Levels = 4, Leaf Size = 10, Predictor Bins = 50, Bin Method = Quantile**.

The final model achieved: **Training ASE = 293, Validation ASE = 934, Test ASE = 935**. Since Validation ASE and Test ASE were almost identical, there is strong evidence that the selected configuration maintained its predictive accuracy when applied to the final unseen test partition. The Iteration Plot also showed a rapid reduction in prediction error followed by stabilisation, supporting convergence of the final 100-tree ensemble. Session_Duration remained the most influential predictor, followed by Avg_BPM, Experience_Level, and Fat_Percentage.

Chapter 4: Comprehensive Model Evaluation

The 4 models were compared using Validation ASE as the primary selection criterion, because model settings were selected using validation data rather than training or test performance. The same data partition was used across all models to ensure a fair comparison. Test ASE was then used as an independent final check of generalisation.



SAS Viya selected **Gradient Boosting** as the champion model. Its **Validation ASE of 934** was the **lowest** among all 4 models, outperforming **Linear Regression (1461.68)**, **Random Forest (3637.52)**, and **Decision Tree (5184.48)**.

This represents an approximately 36% reduction in Validation ASE compared with Linear Regression, the second-best model.

Linear Regression performed strongly and achieved a high R^2 of 0.9852, indicating that much of the variation in Calories_Burned could be explained using the selected predictors. However, the curved residual pattern suggests that some systematic nonlinear structure remained unexplained, limiting its suitability as the final model despite its low prediction error.

Decision Tree produced the weakest overall performance. Although tuning reduced Validation ASE from approximately 7.6k to 5.2k and Test ASE from approximately 7.4k to 6.1k, its final Test ASE remained noticeably higher than Validation ASE. This indicates less stable generalisation and suggests that a single tree could not adequately capture the complexity of the predictor-target relationships.

Random Forest substantially improved on the single Decision Tree, reducing Validation ASE to approximately 3.6k and Test ASE to 3.9k. The relatively small validation-test gap indicates stable generalisation, showing the benefit of combining multiple trees. However, its prediction error remained considerably higher than both Gradient Boosting and Linear Regression.

Gradient Boosting provided the strongest balance of accuracy and generalisation. Its Validation ASE (934) and Test ASE (935) differed by only approximately 1, showing that performance on the validation data was almost fully maintained on unseen test observations. Although its much lower Training ASE (293) suggests that the model fitted the training data more closely, there was little evidence that this flexibility harmed unseen-data performance because validation and test errors remained nearly identical. This supports the model's ability to capture complex relationships without a substantial loss of predictive performance on unseen observations.

Overall, the models revealed a clear trade-off between interpretability and predictive accuracy. Although Linear Regression offered greater interpretability, its residual pattern suggested incomplete linear fit, while Decision Tree and Random Forest produced higher prediction error. For the objective of accurately predicting Calories_Burned, predictive accuracy and generalisation were prioritised, making Gradient Boosting the most appropriate final model.

Chapter 5: Actionable Insights and Recommendations

The recommendations below integrate evidence from the exploratory visualisations, correlation analysis, and SAS Viya variable-importance results to explain how different factors affect Calories_Burned.

Chapter 5.1: Prioritise Session Duration in Calorie Estimation

Session_Duration (hours) showed the strongest and most consistent relationship with Calories_Burned. The scatter plot showed a clear positive pattern, while the correlation analysis produced $r = 0.908$, the strongest correlation with the target.

This was reinforced by SAS Viya, where Session_Duration ranked as the most important predictor in Decision Tree, Random Forest, and Gradient Boosting. Its consistent dominance across different tree-based models indicates that its predictive contribution is robust rather than dependent on a single modelling technique. The gym should therefore prioritise accurate recording of workout duration when estimating calorie expenditure. Incorrect or incomplete session-duration records are likely to materially affect predicted calories because the models consistently relied heavily on this variable.

Chapter 5.2: Combine Average Heart Rate with Session Duration

The Average BPM vs Calories Burned scatter plot showed a positive relationship, although its linear correlation with Calories_Burned was weaker at $r = 0.343$.

However, SAS Viya showed that Avg_BPM remained an important predictor when considered jointly with other variables. It ranked second in both the Decision Tree and Gradient Boosting models, and remained influential in Random Forest, ranking third. The contrast between its modest standalone correlation and stronger model importance suggests that Avg_BPM contributes through multivariable or nonlinear relationships that are not fully captured by pairwise correlation alone. The gym should therefore combine average heart rate with session duration and other workout characteristics rather than estimating calorie expenditure from heart rate independently.

Chapter 5.3: Incorporate Experience Level as a Supporting Predictor

The Average Calories Burned by Experience Level bar chart showed a substantial increase from 726.37 calories for Level 1 to 1265.34 calories for Level 3.

Experience_Level also had a strong positive correlation with Calories_Burned ($r = 0.694$) and ranked among the more important predictors in the Random Forest and Gradient Boosting models, showing that it continued to contribute predictive value when considered together with other variables.

First column name T String	Second column name T String	Correlation value Number (Float)
Workout_Frequency (days/week)	Experience_Level	0.837
Session_Duration (hours)	Experience_Level	0.765
Calories_Burned	Workout_Frequency (days/week)	0.576

Workout_Frequency also showed a moderate positive correlation with Calories_Burned ($r = 0.576$). However, Experience_Level was also strongly correlated with Workout_Frequency ($r = 0.837$) and Session_Duration ($r = 0.765$). This suggests that part of its association with Calories_Burned may overlap with differences in workout behaviour across experience groups, so the relationship should not be interpreted as directly causal. The gym can therefore use experience level as a supporting segmentation variable when estimating expected calorie expenditure, but it should be combined with actual workout characteristics such as session duration and heart rate rather than used independently.

Chapter 5.4: Treat Body Composition and Demographics as Supporting Factors

Fat_Percentage showed a moderately strong negative correlation with Calories_Burned ($r = -0.598$) and ranked among the more important predictors in the ensemble models. This suggests that body composition contributes useful predictive information beyond the main workout variables, particularly when considered together with other predictors. In contrast, BMI showed little standalone relationship with Calories_Burned. Its scatter plot showed weak separation and its correlation was only $r = 0.060$, indicating that BMI alone provides limited information about calorie expenditure. The Age Group bar chart also showed a gradual decrease in average calorie expenditure, from 950.50 calories for Young Adults to 831.73 calories for Older Adults, although the differences were smaller than those observed across experience levels.

Taken together, the results show that body composition and demographic factors add incremental predictive value, but they do not show the same strength and consistency as direct workout measures such as Session_Duration and Avg_BPM. The gym should therefore use these variables to refine and personalise calorie estimates, while keeping workout-related variables as the main drivers of prediction.

Chapter 5.5: Use Gradient Boosting for Individual Calorie Prediction

Gradient Boosting is recommended as the preferred predictive model because it achieved the lowest Validation ASE of 934 and an almost identical Test ASE of 935, indicating both strong predictive accuracy and stable performance on unseen observations. The model can combine the effects of Session_Duration, Avg_BPM, Experience_Level, Fat_Percentage, and the remaining predictors simultaneously. This is preferable to estimating calories from any single factor as the analysis showed that calorie expenditure is influenced by multiple workout, physiological, and member characteristics.

The gym can therefore use Gradient Boosting to generate individual calorie estimates while using SAS Viya variable-importance rankings to explain which factors contribute most strongly at the overall model level. Predicted calorie values should still be treated as estimates rather than exact physiological measurements. Model performance should be reassessed as new data become available to ensure that prediction accuracy remains stable if member characteristics or workout patterns change.

Chapter 6: Conclusion

The analysis identified Session_Duration as the most consistent predictor of Calories_Burned, supported by the exploratory visualisation, the strongest correlation ($r = 0.908$) and its top variable-importance ranking across the 3 tree-based models. Avg_BPM, Experience_Level, and Fat_Percentage also contributed useful predictive information, while BMI and demographic factors were less influential on their own.

Overall, the results show that calorie expenditure is best predicted using a combination of workout, physiological, and member characteristics rather than any single factor. Among the 4 models evaluated, Gradient Boosting achieved the strongest and most stable predictive performance, making it the recommended model for estimating Calories_Burned.